EXPERIMENT REPORT · TRAINING-FREE ACTIVE SENSING

Active Image Sensing: Sequential Candidate Updates

We compare old-candidate guidance, acquisition weights, and reconstruction weights under a frozen prior, with adaptive tempering and full-history guidance.

Experimental report · Concise English edition

> **Scope.** Main results use ordinary CelebA test images and a frozen CelebA-HQ256 Rectified Flow prior. **ADS-alg/RF-prior** adapts ADS to this prior; it is not a reproduction of the original DDIM model. Earlier CelebAMask-HQ #20162 diagnostics are reported separately.

1. Experimental setup

Test set	Five CelebA test images: 187000, 183258, 200793, 189467, 193037; seed 6101.
Measurements and prior	128×128 RGB; direct, nonoverlapping 4×4 pixel blocks, one new block per round. Frozen native256 Rectified Flow prior with 2×2 average pooling D_2; $A=M D_2$. No low-resolution base image.
Initial → total blocks	50→100, 100→200, 250→500; independent groups. Spatially distributed initial blocks count toward each budget.
Candidate generation	K=16; 128 steps per update. Fresh Gaussian initialization with all-history + old-candidate clean-image guidance for U/W; current FlowDPS-style correction and stochastic recombination.
Noise and proximity	$\sigma=0.05$ on $[-1,1]$; $\tau=1$. Energies use summed squared distances.

Controls and evaluation. U/W/From-scratch/Random share initialization, round-paired generation inputs, and position-fixed sensor noise; ADS-RF has its own initialization/randomness. Select before observing; GT is evaluation-only. Compute MSE of the reconstructed mean and PSNR/SSIM from unclipped arrays, averaging images equally; update averages exclude initialization. No test-based tuning or training; prior-training overlap is unverified.

2. Operations used by each method

Method	Acquisition		Generation guidance		Candidate weight update		Reconstruction	
	① Uniform max. disagreement	② Recursive max. disagreement	All history + old candi-date	All history only	Fixed 1/K	Gaussian-product recursion (β)	① Uniform mean	② Weighted mean
UU	✔	—	✔	—	✔	—	✔	—
UW	✔	—	✔	—	—	✔	—	✔
WU	—	✔	✔	—	—	✔	✔	—
WW	—	✔	✔	—	—	✔	—	✔
From-scratch	✔	—	—	✔	✔	—	✔	—

Fixed 1/K means equal candidate weights. **Gaussian-product recursion (β)** uses the old weight and two Gaussian factors, followed by tempering and normalization (Section 3); it is heuristic, not exact Bayes.

✔ marks required operations. **The first letter specifies acquisition weights; the second specifies reconstruction weights.** U uses 1/K; W uses recursively updated weights. WU therefore updates weights for acquisition but uses 1/K for reconstruction. U is not random acquisition.

Notation follows the Math Note (Eqs. 1–3)

Following the Math Note, $g_t^{(i)}$ is candidate i after update t and $w_t^{(i)}$ its normalized weight. K=16; $H_t=((A_1,y_1),\ldots,(A_t,y_t))$; t_0 is the initial budget. Candidates are native256, and $\mathcal{A}_{t-1}$ contains operators A=M D_2 for unobserved 4×4 blocks.

Acquisition: use the previous candidates (Math Note Eq. 3)

$$A_t \in \arg\max_{A \in \mathcal{A}_{t-1}} \sum_{i=1}^{K} w_{t-1}^{(i)} \left\| A g_{t-1}^{(i)} - \sum_{j=1}^{K} w_{t-1}^{(j)} A g_{t-1}^{(j)} \right\|_2^2$$

② WU/WW: use recursive weights. **① UU/UW/From-scratch:** substitute 1/K in both places. Select A_t, observe y_t, generate new candidates, then update weights where needed.

Reconstruction: average the current candidates (Math Note Eq. 2)

$$\hat{f}_{t,\text{uniform}} = \frac{1}{K} \sum_{i=1}^{K} g_t^{(i)}, \qquad \hat{f}_{t,\text{weighted}} = \sum_{i=1}^{K} w_t^{(i)} g_t^{(i)}$$

① UU/WU/From-scratch: left expression. **② UW/WW:** right expression. Evaluate and display $D_2\hat{f}_t$ at task128 resolution.

Trajectory sharing. UU/UW share U; WU/WW share W. U saves recursive weights for UW, but they never affect UU, which can omit this calculation when run alone. WU requires them for its next acquisition. The update formula is shared; weights can differ across trajectories.

Generation and baselines. Four combinations use fresh-noise, full-history + old-candidate guidance; weights do not change guidance or resample candidates. From-scratch removes old guidance; Random also replaces acquisition with random selection. ADS-RF uses its own entropy criterion, a single 600-step trajectory, and uniform reconstruction without our weight recursion.

3. Two changes relative to the Math Note

Change 1: adaptive β to prevent weight collapse

Math Note Eqs. 7–8: old weight × new-measurement Gaussian × old/new-candidate Gaussian.

$$\widetilde{w}_t^{(i)} = w_{t-1}^{(i)}\, \mathcal{N}\big(y_t; A_t g_t^{(i)}, \sigma^2 I_m\big)\, \mathcal{N}\big(g_t^{(i)}; g_{t-1}^{(i)}, \tau^2 I_d\big)$$

Normalize this product. A_t, y_t refer only to the newest block; σ=0.05, τ=1, m=48, d=256×256×3. Initialize recursive weights at $w_{t_0}^{(i)}$=1/K.

Our change: temper the entire product by β before normalization.

$$w_t^{(i)}(\beta) = \frac{(\widetilde{w}_t^{(i)})^\beta}{\sum_{j=1}^{K} (\widetilde{w}_t^{(j)})^\beta}, \qquad \mathrm{ESS}(\beta) = \frac{1}{\sum_{i=1}^{K} (w_t^{(i)}(\beta))^2}$$

$$\beta_t = \max\{\beta \in [0,1] : \mathrm{ESS}(\beta) \geq 8\}, \qquad w_t^{(i)} = w_t^{(i)}(\beta_t)$$

β=1 recovers the original rule; otherwise use bisection to enforce ESS≥8. β scales the entire product, including the old weight; β=0 is uniform. Retain log weights and use log-sum-exp. β uses no GT, and this is not formal SMC. Generation uses full history; measurement scoring uses only the newest block.

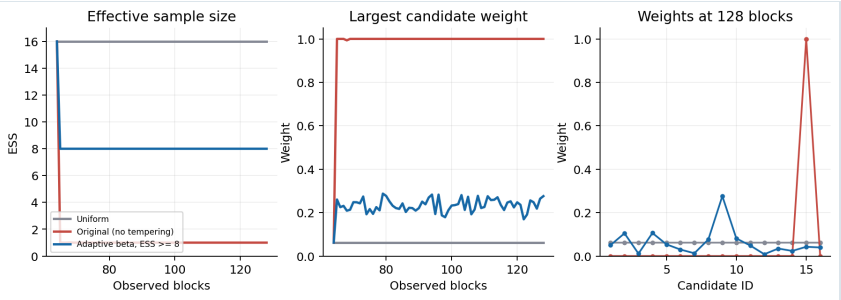


Figure 1. Separate CelebAMask-HQ #20162 diagnostic: 64→128 blocks, K=16. All variants reuse identical candidates/history and restart uniformly. Original weights end at ESS=1 and maximum weight=1; adaptive β ends at ESS=8 and maximum weight ≈0.276.

Reconstruction weights on fixed candidates	Final ESS	Max. weight	Final MSE	Update-mean MSE
Uniform	16.00	0.062	0.005077	0.010824
Original recursion, no β tempering	1.00	1.000	0.006679	0.013775
Adaptive β, ESS≥8	8.00	0.276	0.005304	0.010475

Stability is not accuracy. Adaptive β prevents collapse but ends at MSE 0.005304 versus 0.005077 for uniform weights, despite a slightly better update mean. This offline diagnostic does not evaluate an untempered active W policy.

Change 2: guide generation with the full measurement history

The Math Note uses “old candidate + newest measurement.” We retain old-candidate guidance but explicitly include all acquired measurements in every update, as already done at initialization.

An approximate old image cannot guarantee historical consistency. New-only guidance leaves old regions unconstrained; proximity alone permits errors to accumulate. “Forgetting” here means increasing historical observation error, not changes to trained parameters.

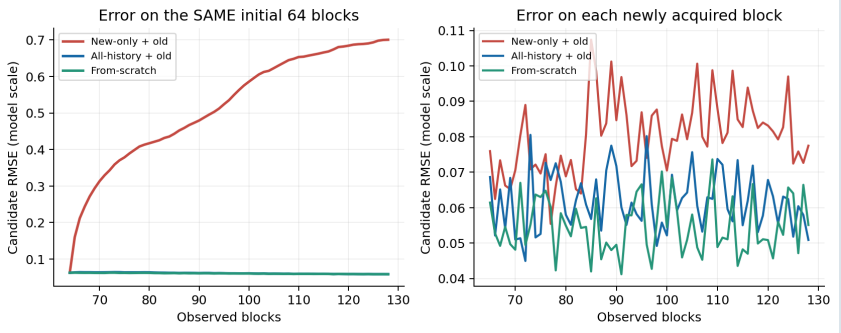


Figure 2a. #20162: candidate RMSE on the fixed initial 64 blocks (left) and each newest block (right; locations may differ). Initial-block RMSE rises from 0.0627 to 0.7004 with new-only guidance; all-history ends at 0.0588.

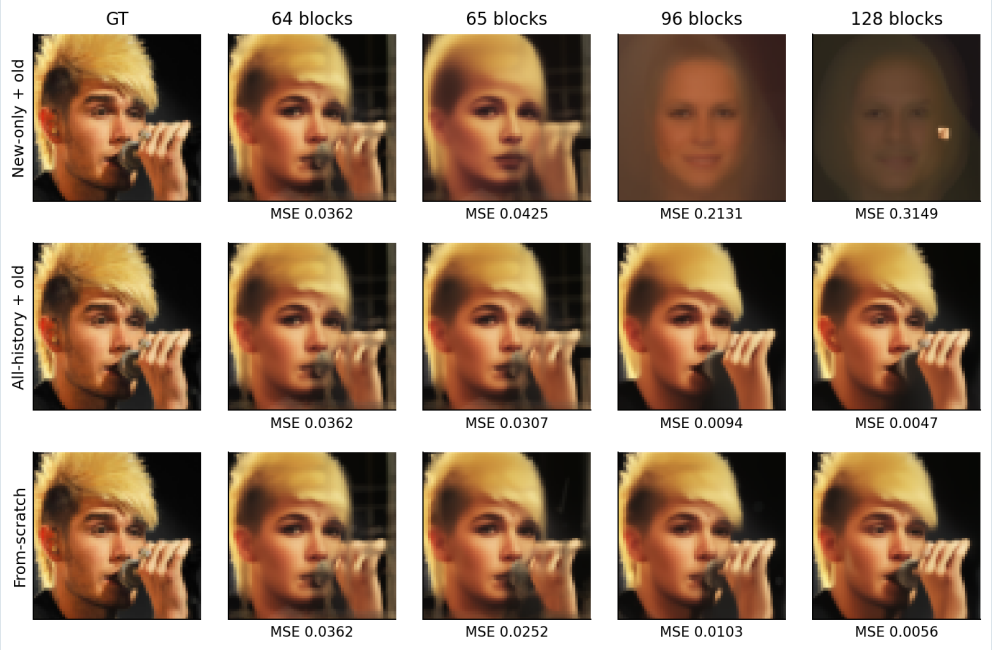


Figure 2b. Uniform means and GT from the same diagnostic: CelebAMask-HQ #20162, task64/native256, 64→128 blocks of 4×4, K=16, 128 steps, σ=0.05, τ=1. MSE uses raw arrays; display clipping only. Separate from the main five-image task128 study.

Diagnostic variant	Initial 64-block candidate RMSE ↓	Newest-block candidate RMSE ↓	Reconstruction MSE ↓
New-only + old	0.7004	0.0775	0.314902
All-history + old	0.0588	0.0509	0.004746
From-scratch	0.0586	0.0551	0.005571

> **Evidence scope.** Initialization and random inputs are paired. At the shared block-65 history, initial-block RMSE is 0.1604 (new-only) versus 0.0637 (all-history). Later active paths may diverge: final differences include acquisition feedback, not only guidance. This is not a full fixed-history causal ablation or a universal claim.

Implementation distinction. Guidance uses all acquired measurements; weight scoring uses the newest block plus full-native-image proximity. FlowDPS-style corrections act on predicted clean images, rather than literally applying the Math Note's noisy-state guidance equation. Unobserved GT is never used.

4. Experimental results

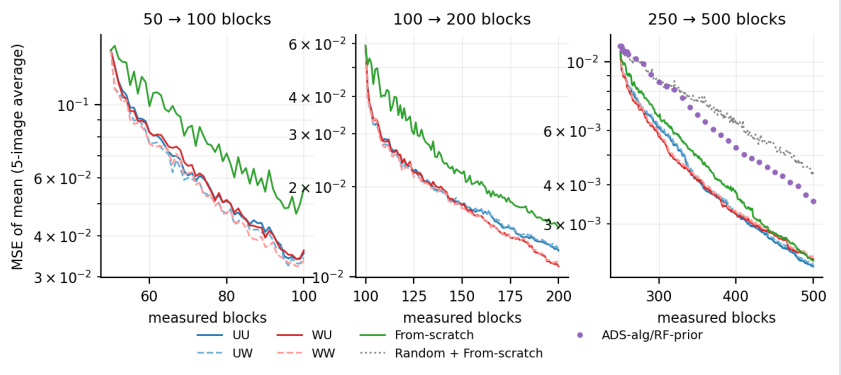


Figure 3. Five-image mean MSE, log scale; three independently initialized budget groups. ADS-RF appears only at its 35 evaluated completion budgets, not at every round.

Old-candidate guidance. UU reduces final mean MSE versus active From-scratch by 36.3%, 16.2%, and 4.2% across the three groups (5/5 images improve in each). Different active paths mean these are complete-method effects.

Weights. Weighted reconstruction helps on average at 50→100. WU is best at the 100→200 endpoint. At 250→500, UU is best finally and WU slightly better during acquisition; UW/WW worsen both final and update-mean MSE versus their uniform counterparts.

Controls. At 250→500, active From-scratch beats random acquisition by 47.5% in final mean MSE; UU beats ADS-RF by 38.4% (4/5 images). ADS-RF differs in both sampler and acquisition criterion.

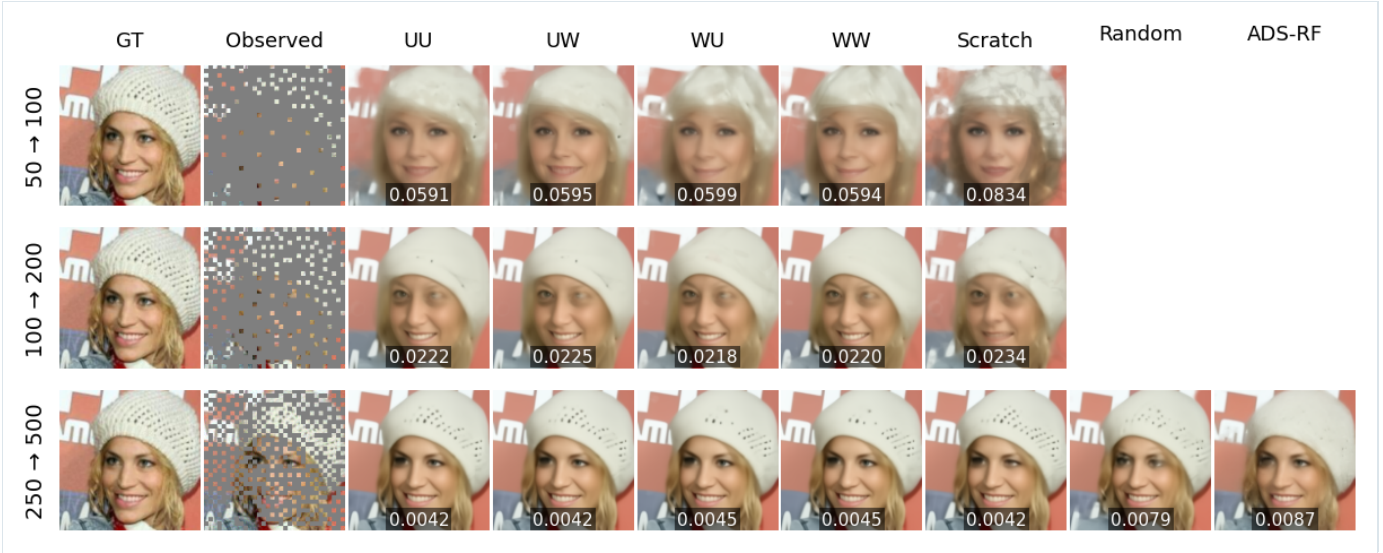


Figure 4. #193037 across budgets, the largest final UU error among the five cases. Observed shows UU's noisy measurements; gray denotes unobserved pixels. Insets: image MSE. Random and ADS-RF run only at 250→500.

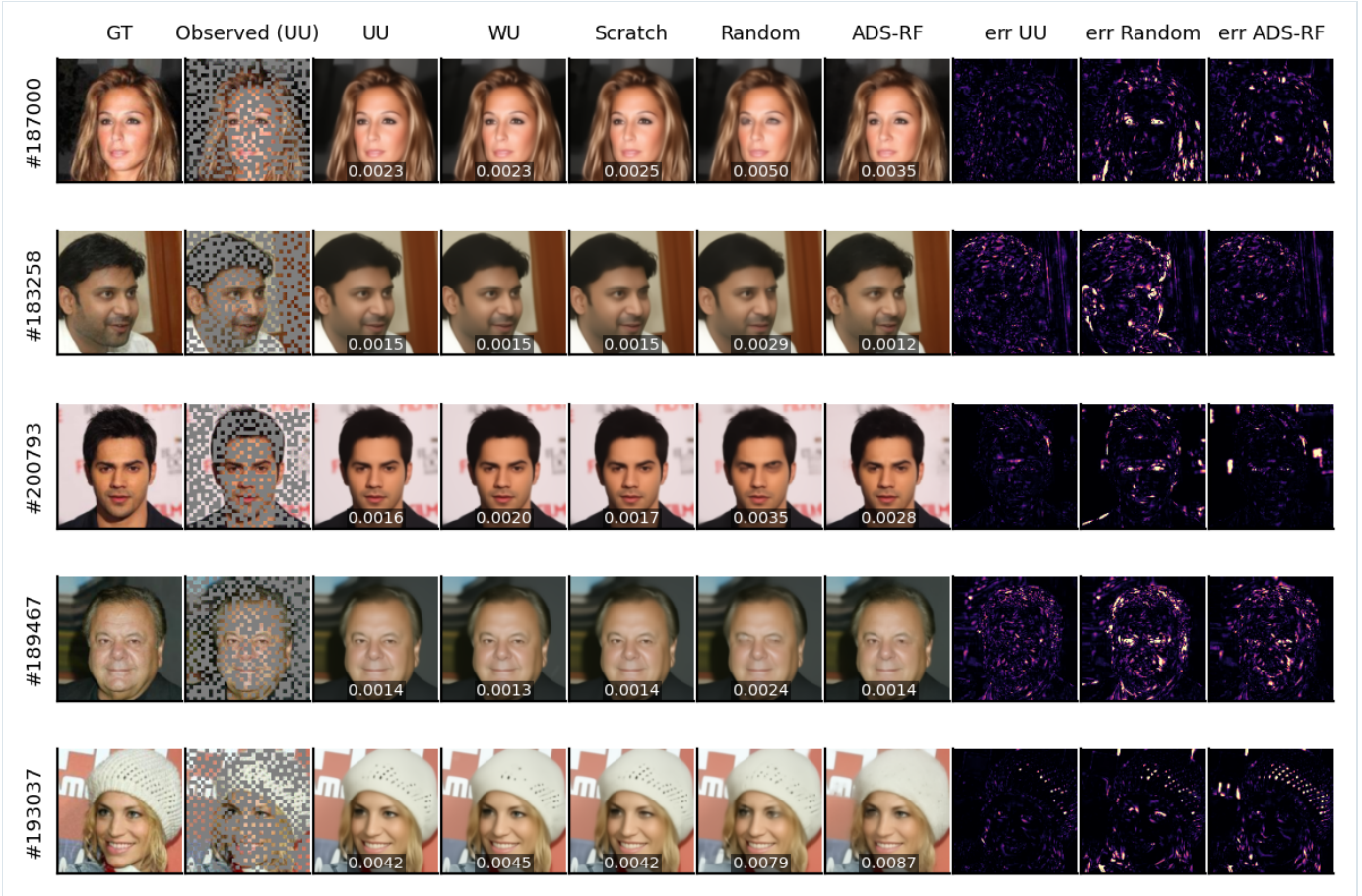


Figure 5. All five images at 500 blocks: GT, UU observations, reconstructions, and RGB-mean squared-error maps for UU/Random/ADS-RF. Maps share a scale within each row. Metrics are unclipped; images are clipped for display.

5. Computational cost

Method	50→100 50 updates	100→200 100 updates	250→500 250 updates
U trajectory (UU/UW)	12.3 min	24.6 min	62.5 min
W trajectory (WU/WW)	12.3 min	25.0 min	62.8 min
From-scratch	12.4 min	24.9 min	62.0 min
Random + From-scratch	Not run	Not run	61.5 min
ADS-alg/RF-prior	Not run	Not run	135 s main trajectory + 62.9 min completion branches

Mean sampling time per image; one NVIDIA GH200 per job. U/W/From-scratch use 16 candidates and 128 steps per update. Initialization, queueing, loading, and evaluation are excluded. Shared reconstruction views need no extra generation.

ADS-RF’s 600-step main trajectory gives its final result; approximately 63 additional minutes produce intermediate-budget reconstructions. Our method still resamples each round and shows no sampling speedup over From-scratch.

Experimental report · Five images, one fixed seed per image. Conclusions are limited to this task and configuration.